

TORINO BRIC DELLA CROCE, Italy

WMO: 160610

Lat: 45.034N

Lon: 7.733E

Elev: 2329

StdP: 13.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	26.4	-5.5	4.5	36.6	0.0	6.0	37.6	26.4	45.4	22.6	44.6	3.0	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	82.8	69.4	80.8	68.6	78.8	67.6	74.1	78.5	72.1	77.0	70.4	75.8	2.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	133.3	76.4	70.7	123.3	75.0	68.6	114.5	73.6	39.4	78.9	37.6	77.6	36.0	76.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	15.4	12.1	DB	21.7	86.5	3.7	2.6	19.0	88.4	16.8	89.9	14.7	91.3	12.0	93.2	
			WB	19.0	75.8	3.2	3.1	16.7	78.0	14.8	79.9	13.0	81.6	10.7	83.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	37.6	39.1	46.6	52.1	59.4	66.7	70.9	70.1	62.9	54.7	45.1	38.9
	DBStd	13.31	6.24	6.76	7.55	6.78	6.28	6.00	5.01	5.53	5.31	5.94	5.83	6.04
	HDD50	1451	390	310	155	53	6	0	0	0	0	25	166	346
	HDD65	4636	850	726	570	388	194	50	13	17	100	321	598	809
	CDD50	2821	4	4	50	117	297	502	648	622	387	170	18	2
	CDD65	534	0	0	0	2	21	102	196	174	37	2	0	0
	CDH74	2105	0	0	0	2	43	399	866	744	49	2	0	0
	CDH80	291	0	0	0	0	2	55	108	125	1	0	0	0
Wind	WSAvg	3.6	3.4	3.4	4.3	4.5	4.3	3.9	3.8	3.3	3.2	2.4	3.0	3.2
Precipitation	PrecAvg	32.1	1.5	2.1	2.3	3.0	4.3	3.6	1.9	3.3	3.2	3.3	2.4	1.4
	PrecMax	52.7	3.7	10.2	5.6	7.0	10.0	6.9	5.5	7.1	10.8	10.2	6.8	5.6
	PrecMin	18.5	0.0	0.1	0.0	0.0	0.4	1.3	0.4	0.0	0.1	0.0	0.1	0.0
	PrecStd	8.7	1.1	2.7	1.6	2.0	2.5	1.5	1.3	1.9	2.8	2.8	2.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	59.0	67.8	72.0	78.3	84.6	85.7	87.4	78.2	71.1	62.5	57.3
		MCWB	45.1	45.3	51.8	57.6	63.9	69.4	71.9	71.3	65.5	59.2	50.1	41.7
	2%	DB	53.8	54.0	63.0	67.9	74.8	81.0	82.7	83.3	75.3	66.4	57.8	52.1
		MCWB	41.2	42.1	49.1	55.1	62.0	68.1	69.5	70.0	64.3	58.0	49.6	39.5
	5%	DB	48.5	50.4	59.6	64.6	71.6	78.6	80.7	80.6	73.0	64.3	54.3	48.6
		MCWB	39.3	40.6	47.9	53.5	60.2	67.2	68.7	69.0	63.1	57.7	47.3	39.2
	10%	DB	45.5	47.9	56.9	61.9	68.4	75.6	78.8	78.1	70.2	62.3	52.1	46.5
		MCWB	37.7	39.8	46.6	52.0	58.3	65.6	67.6	68.0	61.5	57.1	46.9	38.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	48.2	55.5	61.5	68.8	74.6	76.8	77.1	70.1	64.1	55.3	47.6
		MCDB	56.4	52.3	61.7	67.0	74.3	79.2	81.0	80.7	74.1	66.3	59.4	49.7
	2%	WB	43.9	45.5	52.5	58.0	64.7	72.0	74.1	74.2	68.0	61.7	52.3	44.7
		MCDB	49.3	49.7	58.5	64.4	70.8	77.0	78.4	78.4	72.3	63.8	54.8	47.0
	5%	WB	41.7	43.6	50.2	55.4	62.4	69.9	72.1	72.1	66.3	60.2	50.6	42.7
		MCDB	45.5	47.5	56.5	62.4	68.8	75.0	76.9	76.9	70.1	62.4	52.5	45.3
	10%	WB	40.0	41.7	48.4	53.3	60.5	67.9	70.4	70.3	64.4	58.5	49.1	41.0
		MCDB	43.9	46.0	54.4	59.8	66.2	74.1	75.9	75.7	68.4	61.0	50.9	43.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.5	11.1	11.6	12.0	12.6	12.9	11.8	10.2	7.5	6.7	6.8
		MCDBR	11.4	11.5	13.8	14.2	14.3	13.8	13.6	13.1	12.2	9.6	8.7	9.7
		MCWBR	7.9	8.2	9.0	8.7	9.1	8.8	8.9	8.7	7.8	7.0	7.4	7.1
	5% WB	MCDBR	8.9	9.7	12.3	12.7	12.7	12.5	12.3	11.7	10.5	8.0	6.4	7.1
		MCWBR	7.4	7.8	8.8	8.8	9.0	9.2	9.3	8.6	7.2	6.4	6.4	6.7
Clear-Sky Solar Irradiance	taub		0.291	0.313	0.351	0.392	0.396	0.410	0.399	0.394	0.382	0.364	0.321	0.286
	taud		2.478	2.422	2.344	2.246	2.283	2.282	2.312	2.333	2.342	2.383	2.467	2.500
	Ebn at Noon		264	277	279	275	277	272	274	272	265	253	248	254
	Edh at Noon		22	28	34	41	41	41	40	37	34	29	22	19
All-Sky Solar Radiation	RadAvg		521	779	1145	1415	1679	1962	2028	1754	1339	844	515	440
	RadStd		61	100	141	156	144	138	133	103	100	108	77	56

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (54 neighbors)	+0.56	N/A	N/A	+1.13	N/A	N/A	N/A	N/A	+108	+32	

Nomenclature: See separate page